

# HARIS WAHEED BHATTI

Nationality: Pakistani | Date of birth: 25/12/2001

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**Work Authorisation:** Pakistani national residing in Slovenia on a temporary student residence permit with full-time work rights; eligible for a post-study job-search residence permit from September 2026.

## PROFILE

BIM and digital construction specialist completing a European Master's in Building Information Modelling at the University of Ljubljana, building on a Bachelor's in Civil Engineering from NUST, Pakistan. Experienced across IFC-to-graph-database pipelines and blockchain-based construction document management. Seeking roles such as **Information Manager**, **BIM Specialist**, **AI in BIM**, or **IFC Expert** in Slovenia.

## SKILLS

**BIM & AEC:** IFC, IfcOpenShell, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, WebIFC  
**Programming & Data:** Python, TypeScript, Solidity, Neo4j, Cypher, knowledge graphs, Git, LaTeX  
**AI / LLM:** agentic systems, Claude Agent SDK, MCP, LLM evaluation and benchmarking  
**Languages:** Urdu (native), English (fluent)

## EDUCATION

- European Master in Building Information Modelling (BIM A+)** 10/2024 – 07/2026  
*University of Ljubljana, UL FGG* Slovenia
- Field of study: Information Management in Construction
  - Thesis:** *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches.* Supervisor: Prof. Žiga Turk
- BE Civil Engineering** 09/2020 – 06/2024  
*National University of Science and Technology* Pakistan
- Final grade: 3.25/4.00
  - Thesis:** *Enhancing Construction Transparency and Dispute Resolution Through Blockchain-Based BIM Models* – developed a **Solidity** smart contract to digitally sign and manage construction documents, with off-chain storage linked to the blockchain for immutability and transparency
  - Funded by a PKR 100,000 grant from the Pakistan Engineering Council; designed to streamline dispute resolution for construction stakeholders
  - Link: [bit.ly/49dUTmK](https://bit.ly/49dUTmK) | Contract: [sepolia.arbiscan.io/address/0x8535...c867](https://sepolia.arbiscan.io/address/0x8535...c867)

## PROJECTS

- BIMGraphExplorer** | *Neo4j, Cypher, Python, IfcOpenShell, MCP* 2026
- Published `ifctoneo4j`, a Python package converting IFC (IFC2X3, IFC4, IFC4X3) into **Neo4j** property graphs under the Linked Building Data ontologies with no RDF intermediate
  - Built an adaptive querying system exposing four read-only tools over the Model Context Protocol, in which the model profiles the graph schema and issues iterative Cypher queries conditioned on prior results
  - Ran a controlled benchmark against a published direct-IFC baseline at matched model, matched validated judge, and matched scope of 23 IFC datasets (367 questions, 15 projects): 72.0 % versus 82.3 % accuracy, transparency at parity (99.7 % versus 100 %), and outright wins on 5 of 23 datasets
  - Raised accuracy by 33 percentage points on a fixed model and fixed dataset through prompt design alone, lifting source-transparency from 12.5 % to 99.7 %
- Pakistani-Themed Expo BIM Management** | *IFC, Autodesk Construction Cloud* 28/10/2024 – 19/11/2024
- Spearheaded a BIM management project for a Pakistani-themed expo in Guimarães, Portugal, leveraging **IFC** and **Autodesk Construction Cloud** to develop a comprehensive technical design
  - Facilitated collaboration between Architecture and MEP disciplines, creating a federated model that integrated diverse design inputs
  - Conducted clash detection and compliance checks to ensure the model met all information requirements and design standards

## EXPERIENCE

- Contract Management Intern** 08/2023 – 09/2023  
*Engineering and Management Services* Pakistan
- Reviewed construction contract reports and arbitration case documentation under **FIDIC** conditions, and presented findings on FIDIC and arbitration procedures to the internal legal team
- Research Intern (Part-time, Remote – Concurrent with Above)** 07/2023 – 08/2023  
*Southern Illinois University Edwardsville* USA
- Extracted and synthesized findings from academic articles and abstracts to support a faculty-mentored research proposal, applying feedback to refine presentation and analysis technique
  - Supervisor: [linkedin.com/in/saadullah1](https://www.linkedin.com/in/saadullah1)
- Civil Engineering Intern** 07/2023 – 08/2023  
*Fatima Fertilizer Company* Pakistan
- Modeled a 42m x 162m (6,804 m<sup>2</sup>) rock phosphate storage shed independently in 3D from 2D drawings, coordinating with stakeholders to refine the design